

CSS Box Model

1. Content

- The content area is the actual text, image, or other material inside the element.

2. Padding

- Padding is the space between the content and the border.
- It adds inner space.

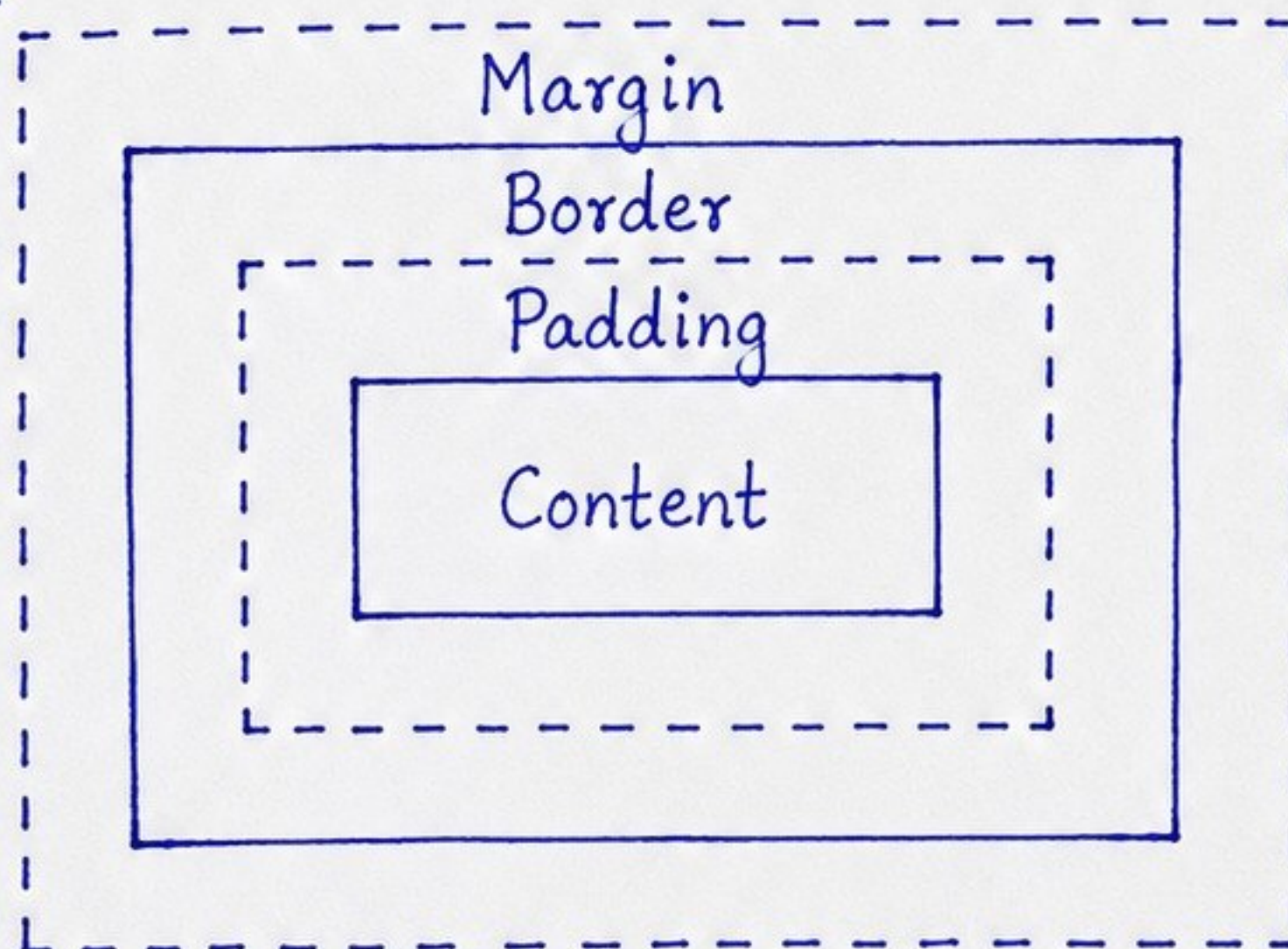
3. Border

- The border wraps around the padding and content.

4. Margin

- Margin is the outer space outside the border.
- It creates distance from other elements.

5. Diagram



Padding

1. padding

- The padding property adds inner space inside an element.
- It creates space between the content and the border.

2. Four sides

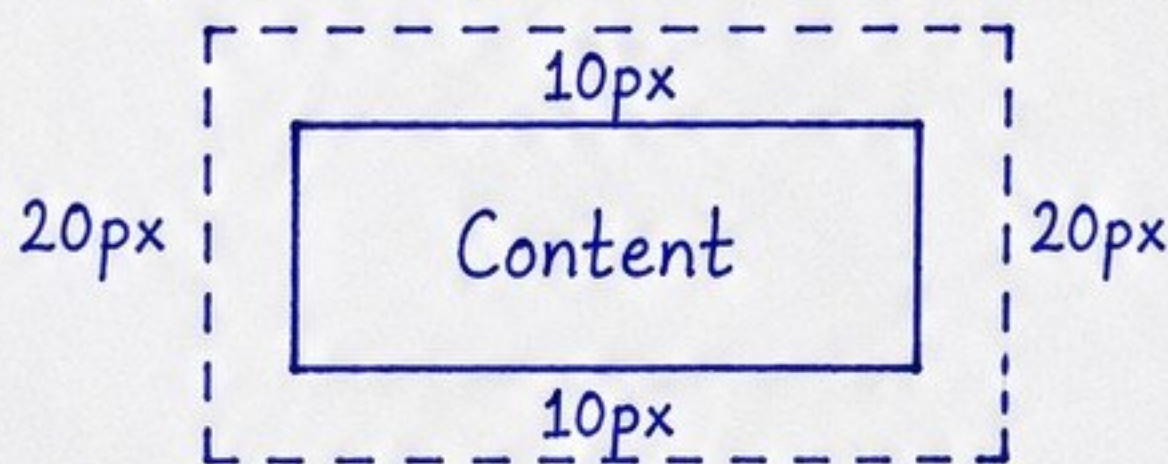
- padding-top
- padding-right
- padding-bottom
- padding-left

3. Shorthand

- padding: 10px;
- padding: 10px 20px;
- padding: 10px 20px 30px 40px;

4. Examples

- padding: 15px; gives equal padding on all sides.
- padding: 10px 20px; means top-bottom 10px and left-right 20px.



Margin

1. margin

- The margin property adds outer space outside an element.
- It creates distance between one element and other elements.

2. Four sides

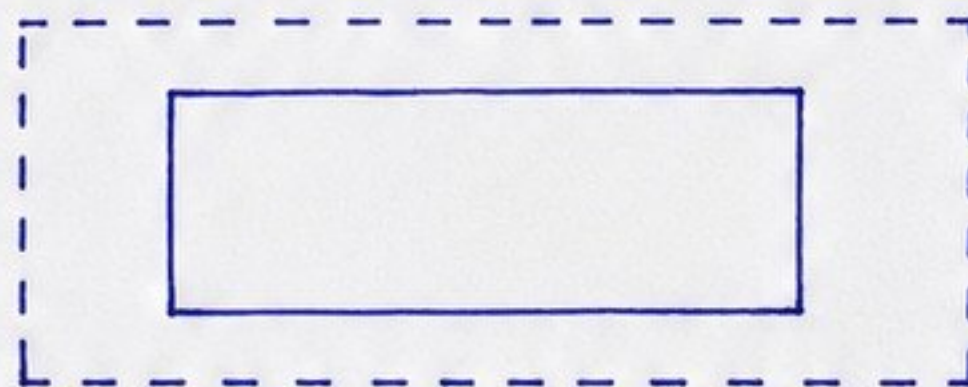
- margin-top
- margin-right
- margin-bottom
- margin-left

3. Shorthand

- margin: 10px;
- margin: 10px 20px;
- margin: 10px 20px 30px 40px;

4. margin: auto

- It is often used to center a block element horizontally.
- Example: width: 300px; margin: auto;



Margin Collapse

1. What it means

- Margin collapse happens when the vertical margins of two block elements touch.
- Instead of adding both margins, the browser usually uses the larger one.

2. Vertical margins

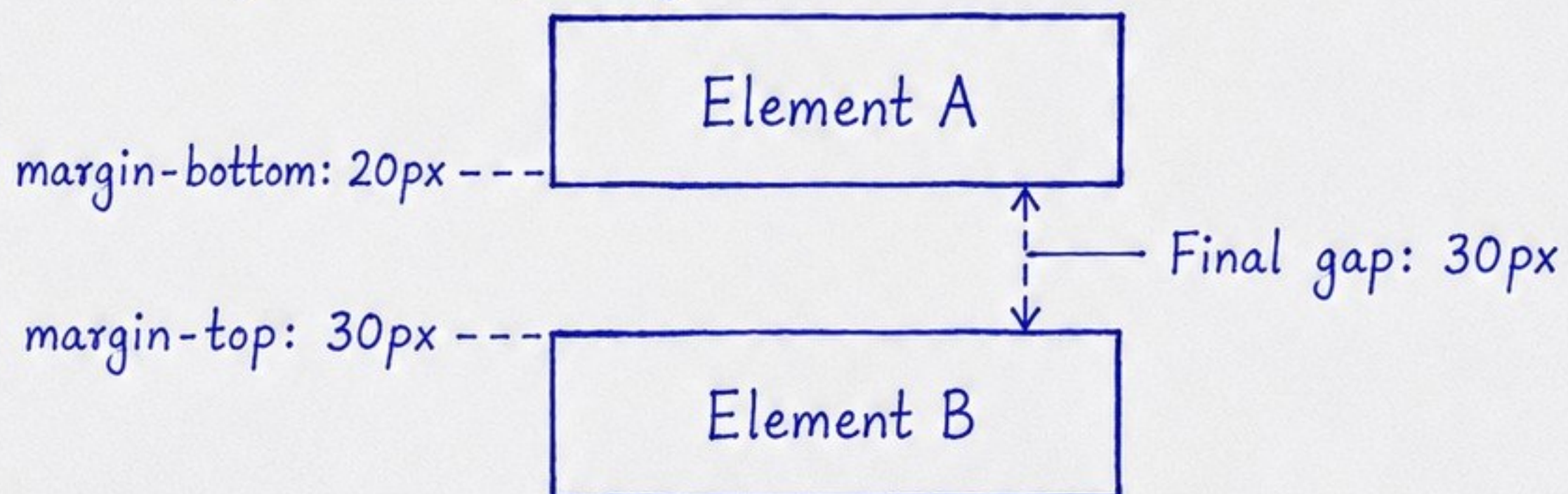
- It mainly affects top and bottom margins.
- Left and right margins do not collapse in the same way.

3. Simple example

- If one element has margin-bottom: 20px and the next has margin-top: 30px,
- the final gap is usually 30px, not 50px.

4. Important note

- Understanding margin collapse helps avoid unexpected spacing.



box-sizing

1. content-box

- This is the default box-sizing value in CSS.
- Width and height apply only to the content area.
- Padding and border are added outside that size.

2. border-box

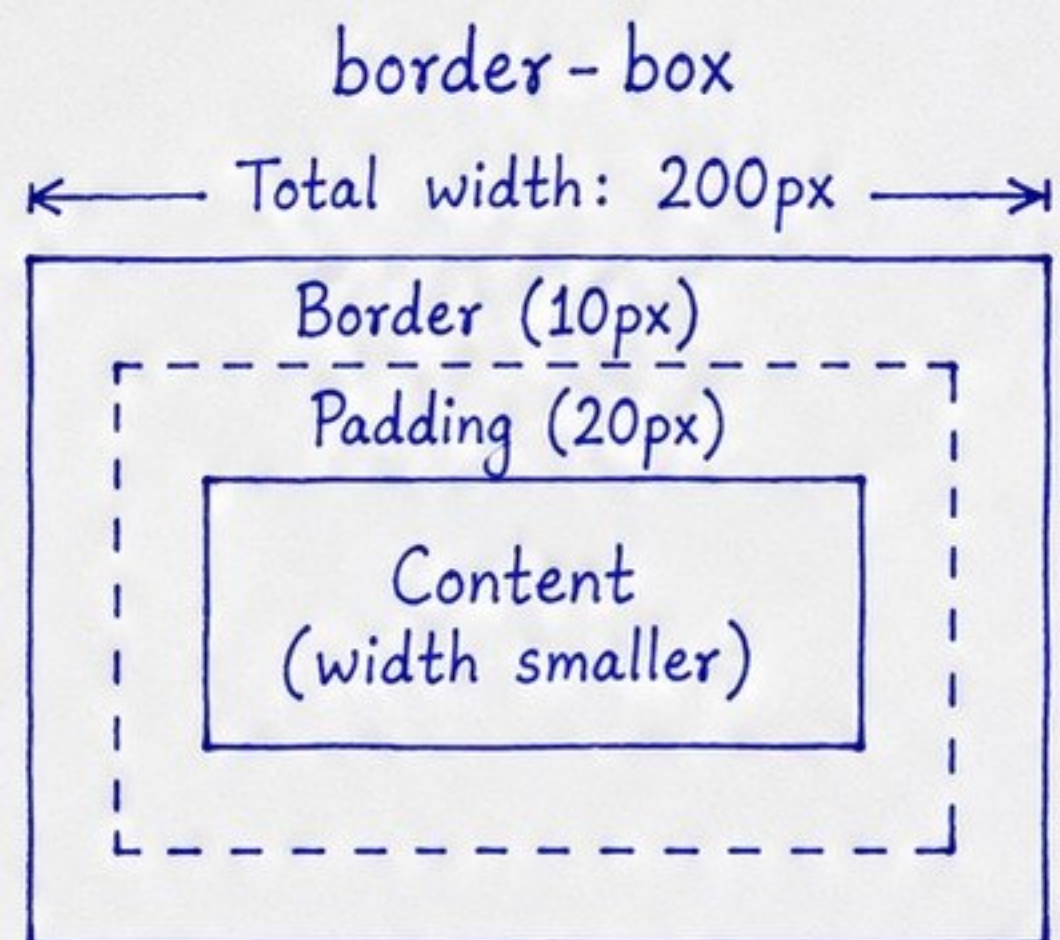
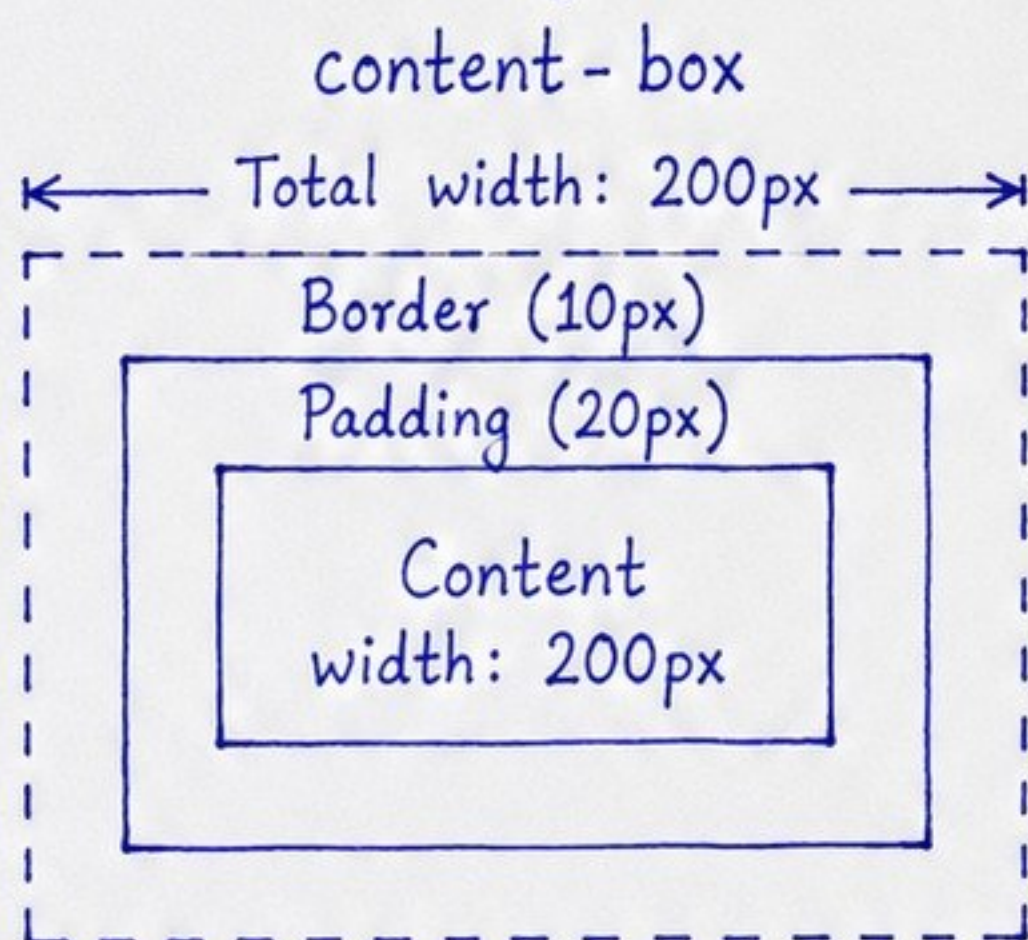
- Width and height include content, padding, and border.
- This makes layout sizing easier.

3. Why border-box is useful

- It gives more predictable element sizes.
- It helps avoid layout problems.

4. Example

- `box-sizing: border-box;`



Text Color and Alignment

1. color

- The color property changes the color of text.
- Example: color: blue;

2. text-align

- The text-align property controls horizontal alignment.

3. Common values

- left
- center
- right
- justify

4. Example

- text-align: center;
- text-align: justify;

5. Note

- Text color and alignment improve readability and presentation.

Text Decoration

1. text-decoration

- The text-decoration property adds or removes decoration from text.

2. Common values

- underline
- overline
- line-through
- none

3. Examples

- text-decoration: underline;
- text-decoration: line-through;
- text-decoration: none;

4. Uses

- underline can highlight links.
- line-through can show deleted or old text.
- none removes the default line from links.

Text Transformation

1. text-transform

- The text-transform property changes the case of text.

2. Common values

- uppercase
- lowercase
- capitalize

3. Examples

- text-transform: uppercase;
- text-transform: lowercase;
- text-transform: capitalize;

4. Note

- It changes how text appears on the page without changing the original HTML text.

Text Spacing

1. letter-spacing

- Controls the space between letters.

Example: `letter-spacing: 2px;`

2. word-spacing

- Controls the space between words.

Example: `word-spacing: 5px;`

3. line-height

- Controls the vertical space between lines of text.

Example: `line-height: 1.6;`

4. text-indent

- Adds space at the beginning of the first line of a paragraph.

Example: `text-indent: 20px;`

5. Note

- These properties improve readability and text appearance.

Font Properties

1. font-family

- It chooses the typeface of the text.
- Example: `font-family: Arial, sans-serif;`

2. font-size

- It controls the size of the text.
- Example: `font-size: 18px;`

3. font-style

- Common values: `normal`, `italic`.

4. font-weight

- Common values: `normal`, `bold`, 100 to 900.

5. Note

- These properties control the appearance and readability of text.